J Insect Behav (2026) 39:24
https://doi.org/10.1007/s10905-026-09919-z

RESEARCH

# Genotype-dependent Food Competition Behaviors in Male Fruit Flies Under Distinct Thermal Conditions

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Received: 10 October 2025 / Revised: 27 May 2026 / Accepted: 30 July 2026


**Abstract** Ambient temperature strongly influences behavioral performance in ectothermic animals. Food-related aggression, as a social interaction centered on a defined and controllable resource, provides an ecologically relevant and experimentally tractable framework for studying conflict. However, how offensive and escape behaviors during such competition vary across thermal conditions and genetic backgrounds remains incompletely understood. Here, we compared food-related competitive behaviors in two laboratory strains of male *Drosophila melanogaster*, *Canton-S* and $w^{1118}$, under two assay temperature conditions. We quantified wing threats, physical contact, and withdrawal behaviors. Overall offensive actions were lower under the cooler assay condition (~18 °C) than under the warmer assay condition (~25 °C), but genotype-related differences were most evident under the cooler condition. Under ~18 °C, *Canton-S* males exhibited higher levels of offensive behavior than $w^{1118}$ males, whereas these genotype differences were less pronounced under ~25 °C. Wing threat displays showed relatively weak genotype differences, whereas physical contact behaviors showed clearer condition-associated and genotype-related variation. Locomotor assays revealed genotype-dependent differences under the cooler condition, indicating that the reduced offensive behavior of $w^{1118}$ males cannot be explained solely by general locomotor suppression. Together, these results suggest that food competition-related behaviors differ between male fruit fly strains across assay conditions, with genotype-related differences being most evident under cooler conditions.

**Keywords** Offensive action · Escape action · Physical contact · Wing threat · *Drosophila melanogaster*

**Supplementary Information** The online version contains supplementary material available at https://doi.org/10.1007/s10905-026-09919-z.

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## Introduction

Animals exhibit diverse behavioral strategies when facing environmental changes. These adaptive behaviors not only are critical for individual survival and reproduction in dynamic environments, but also shape the evolutionary trajectory of species. Variations in environmental adaptability directly influence individual survival, competitive ability, and population stability (Marshall et al. 2017; O'Connell 2020; Rahman and Candolin

Published online: 10 August 2026

2022). Understanding how organisms adjust their behavior under varying environmental conditions is therefore of fundamental importance to behavioral ecology and evolutionary biology.

Temperature variation is a key ecological factor influencing animal behavior. A wide range of animals exhibit pronounced behavioral changes under different thermal conditions—for example, increased aggression in heat-stressed honeybees (Stabentheiner et al. 2007), increased predation efficiency in beetles at higher temperatures (Frank and Bramböck 2016), and temperature-dependent antipredator behaviors in snakes and mammals (Citadini and Navas 2013; Cao et al. 2025). These findings suggest that temperature shifts can influence not only general activity levels but also more specific social and competitive behaviors across taxa. To explore the mechanistic and genetic basis of such temperature-dependent behavioral plasticity, a model organism with well-established tools and experimental tractability is essential. As a genetically tractable model organism, the fruit fly (*Drosophila melanogaster*) provides an ideal system for investigating genotype-specific behavioral plasticity under controlled experimental conditions. In fruit flies, earlier research has indicated that developmental temperature can influence male territorial success, suggesting that thermal environments may modulate competitive interactions between individuals (Zamudio et al. 1995). However, it remains unclear how offensive and escape behaviors during direct resource competition vary under different thermal assay conditions, and whether such patterns differ across genetic backgrounds.

Among the behavioral adaptations influenced by temperature, competitive aggression over food is a tractable model for laboratory investigation. Previous studies have established that fruit flies display aggression and escape behavior when competing for limited food sources (Lim et al. 2014; Asahina 2017; Bath et al. 2021), making this paradigm widely used to study the neural and genetic mechanisms underlying social conflict. Because food competition is sensitive to both internal state and environmental conditions, it provides a suitable context for examining condition-associated variation in competitive behavior. Although laboratory strains have been maintained under stable rearing conditions for decades, their behavioral responses to ambient temperature remain plastic and can provide insights into general principles of thermal modulation in ectotherms (Angilletta et al. 2002; Terblanche et al. 2007).

Moreover, in studies of competitive aggression, the quantification of behavioral indicators is crucial for understanding how fruit flies shift strategies under different environmental conditions. Previous research has shown that different types of aggressive behaviors vary in energetic cost and functional roles (Chen et al. 2002). For instance, physical contact behaviors (such as lunging or boxing) are generally considered high-cost actions, whereas non-contact behaviors like wing threats may incur lower costs and serve as exploratory signals in the early stages of conflict. By simultaneously recording these behavioral components, researchers can more comprehensively characterize how different behavioral categories vary across assay conditions and genetic backgrounds, thereby revealing which components of competitive behavior are most sensitive to environmental and physiological constraints.

As a widely used model organism, *Drosophila melanogaster* has been instrumental in advancing our understanding of neural function and behavioral regulation (Seelig and Jayaraman 2015; Han et al. 2021, 2025). Among the most commonly used laboratory strains are *Canton-S* and $w^{1118}$, which differ significantly in genetic background and physiological traits, making them ideal for comparative studies. *Canton-S*, established from wild flies collected in Ohio in the 1920s, is considered the standard reference strain (Stern 1943; Stern and Schaeffer 1943). In contrast, $w^{1118}$ originated from the *Oregon-R* strain, which was established in the early twentieth century from flies captured in Oregon and is one of the earliest domesticated laboratory lineages (Lindsley 1967). $w^{1118}$ carries a null mutation in the white gene, which leads not only to visual impairment (Colley 2012), but also to differences in locomotion (Colomb and Brembs 2015; Xiao and Qiu 2020), social behavior (Krstic et al. 2013), and neurodegeneration (Goedert and Jakes 2005) compared with *Canton-S*. Importantly, previous studies have shown that white-eyed mutants, including $w^{1118}$, exhibit significantly reduced levels of aggression compared with red-eyed or typical wild-type strains, a phenotype directly relevant to competitive behavior (Hoyer et al. 2008). In addition, prior work has reported that $w^{1118}$ males display reduced behavioral expression across multiple social contexts; for example, both the initiation frequency and display intensity of courtship behavior are lower than those observed in wild-type strains, and responses to social stimuli are also attenuated (Ramin et al. 2014; Xiao et al. 2017). These findings indicate that $w^{1118}$ differs from canonical wild-type strains in the initiation and expression of social interactions. Other studies have further shown that $w^{1118}$ differs from other strains in physiological responses to temperature variation, including thermal tolerance and stress resistance (Myers et al. 2021). Although $w^{1118}$ is not a strict wild-type strain, it has been widely used in behavioral and neurobiological research due to its genetic stability and suitability for transgenic manipulation. Previous studies have indicated that, under standard laboratory conditions, $w^{1118}$ and *Oregon-R* may exhibit similar patterns in certain basic behaviors, such as circadian rhythmicity and general locomotion (Nichols 2007); however, $w^{1118}$ has been reported to differ significantly from *Oregon-R* in courtship-related measures, and white mutants or white genetic backgrounds (including $w^{1118}$) have also been associated with reduced aggressive behavior (Hoyer et al. 2008; Krstic et al. 2013).

In this study, we used *Canton-S* and $w^{1118}$ male flies to compare food competition-related offensive and escape behaviors under two assay temperature conditions. Our goal was to determine whether genotype-related differences in competitive behaviors were evident under cooler and warmer assay conditions. By comparing the performance of the two strains, this study contributes to our understanding of genotype-associated variation in social behavior and behavioral performance in fruit flies.

## Methods

### Fly Strains

*Canton-S* and $w^{1118}$ male fruit flies were cultured in a 25 °C incubator with approximately 50% relative humidity. The incubator was set to a 12:12 light–dark cycle, with lights from 9:00 a.m. to 9:00 p.m. and darkness from 9:00 p.m. to 9:00 a.m. *Canton-S* flies were obtained from the Bloomington Drosophila Stock Center (BDSC, RRID: BDSC_64349), and $w^{1118}$ flies were obtained from the Vienna Drosophila Resource Center (VDRC), with the FlyBase ID: FBgn0003996 (CG2759).

To reduce variation in developmental and rearing conditions, all flies were reared under identical and constant laboratory conditions throughout the experiment. In addition, to better approximate naturally occurring temperature conditions while minimizing temperature fluctuations, behavioral assays were conducted indoors in the laboratory. Given that typical indoor temperatures in the local area during November–December and April–May are approximately 18 °C and 25 °C, respectively, aggression assays were performed during these corresponding periods and ambient temperature was regulated using air conditioning to maintain the target assay temperature conditions. Specifically, "cool" trials were conducted between November and December, with indoor laboratory temperature maintained at 18.0°C ± 0.5 °C, whereas "warm" trials were conducted between April and May, with indoor laboratory temperature maintained at 25.0°C ± 0.5 °C. These temperature ranges have been widely adopted in previous studies on fruit fly physiology and behavior (Nunney and Cheung 1997; Berrigan and Partridge 1997). Throughout behavioral testing, ambient temperature was continuously monitored to ensure that it remained within the specified ranges.

### Tracking Apparatus

Ten *Canton-S* males and ten $w^{1118}$ males were placed together in a transparent circular arena measuring 6 cm in diameter and 1.5 cm in height. A spherical food source approximately 0.5 cm in diameter, prepared from a standard commercial cornmeal–yeast–agar fly medium, was placed at the center of the arena to induce food competition. The arena was covered with a transparent lid to prevent escape. No additional physical spacer, shallow monolayer chamber, or chemical barrier was used to strictly confine flies to a two-dimensional monolayer. This design was chosen because the present assay involved group-level competition around a central spherical food source. Because the food source was approximately 0.5 cm in diameter, an overly shallow chamber could have physically interfered with flies standing on or around the food, occupying the food patch, and engaging in contact or withdrawal behaviors near the food source. However, the 1.5 cm chamber height did not completely prevent flies from walking on the side wall or underside of the lid. Therefore, behavioral events were scored only when the interacting flies and their actions were clearly visible, and ambiguous events caused by vertical displacement, overlap, or poor visibility were excluded from quantitative analysis.

Each behavioral trial lasted 10 min and was conducted under controlled temperature conditions. *Canton-S* and $w^{1118}$ males were distinguished during video analysis by eye color, with *Canton-S* displaying red eyes and $w^{1118}$ displaying white eyes. Details regarding video acquisition, recording parameters, and quality control procedures are described in the Video Recording and Quality Control section.

### Video Recording and Quality Control

All behavioral assays were recorded using a fixed video acquisition setup to ensure consistent image quality and reliable behavioral scoring. Videos were recorded using an iPhone 12 Pro Max (Apple Inc., USA) at a resolution of 1920 × 1080 pixels (1080p) and a frame rate of 30 frames per second. The smartphone was mounted in a fixed position directly above the arena, providing a vertical top-down view throughout the entire recording. Autofocus was enabled prior to recording and remained fixed during each trial. The field of view was not adjusted, zoomed, or repositioned during recordings used for quantitative analysis.

Recordings were conducted inside an experimental incubator with the door open; therefore, the ambient temperature during behavioral recording was identical to the laboratory room temperature. The smartphone was placed on the first shelf of the incubator, and the behavioral arena was positioned on the second shelf below, with a vertical distance of approximately 30 cm between the camera and the arena. A white background was placed beneath the arena to enhance contrast and facilitate visual identification of individuals and behaviors.

Because the arena did not strictly restrict flies to a monolayer, we addressed potential issues related to occlusion and image clarity by scoring behavioral events only when the interacting flies and their actions were clearly visible. Video segments in which behaviors could not be unambiguously identified due to overlap, obstruction, or blur were excluded from scoring. No changes to camera position, focus, or field of view were made during recordings used for behavioral quantification. Supplementary Files 1–9 were prepared solely for illustrative purposes and may include cropped or zoomed views to highlight representative behaviors; these modifications were not applied to videos used for data analysis. A photograph of the complete video recording setup, including the behavioral arena with the central food source, is provided in Supplementary File 10.

### Experimental Design

*Canton-S* and $w^{1118}$ male fruit flies were housed in groups of 10 individuals of the same genotype in standard food vials immediately after eclosion. Previous research has shown that food deprivation can enhance aggressive motivation in fruit flies, especially when competition over food is involved (Edmunds et al. 2021). Our starvation protocol was adapted from this previous study. From days 3 to 5 post-eclosion, at 9:00 p.m., each group was transferred to an empty vial without food for a 12-h starvation period. To prevent desiccation during starvation, the cotton plug of each vial was moistened with water to maintain humidity. Behavioral observations began the following morning at 9:00 a.m. For each trial, 10 *Canton-S* males and 10 $w^{1118}$ males were placed together in the circular arena (Fig. 1). Before transfer to the behavioral arena, flies were briefly immobilized by cold exposure rather than $CO_2$ anesthesia. Groups of flies were kept in empty vials and inserted into an ice bucket for approximately 20 s until movement ceased. Immobilized flies were then gently transferred into the behavioral arena. After all flies were introduced into the arena, they were allowed to recover for 1 min, by which time all flies had resumed locomotor activity. Video recording was then started, and the 10-min behavioral observation period was defined as beginning when the first fly located the food source.

(a) Ten *Canton-S* and ten $w^{1118}$ male fruit flies (20 males per group) were raised to 3–5 days posteclosion (indicated as D3–D5, where "D" denotes "Day") and transferred into empty vials at 9:00 p.m. for 12 h of starvation. The following morning, the flies were subjected to a custom-designed aggression assay for behavioral observation and quantification. (b) The aggression assay consisted of a transparent circular arena (6 in diameter, 1.5 cm in height) covered with a glass lid to prevent the escape of the flies. At the center of the arena, a spherical food source approximately 0.5 cm in diameter

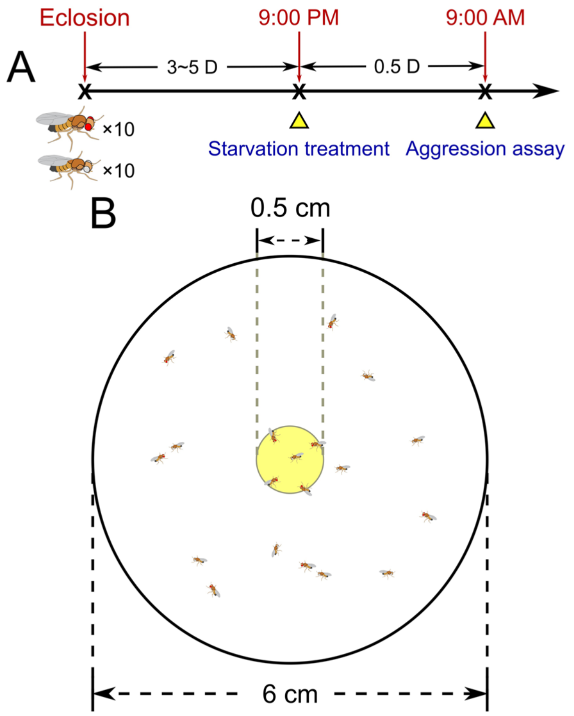


**Fig. 1** Schematic diagram illustrating the experimental design for food competition-induced behaviors in male fruit flies (**a**) experimental timeline and starvation/assay procedure; (**b**) schematic of the aggression assay arena and central food source

(equivalent to approximately two body lengths of a fruit fly) was placed to induce competition. The body size of the flies shown in the schematic is drawn to scale based on measured body lengths (*Canton-S*: mean ± SD = 2.61 ± 0.20; $w^{1118}$: 2.42 ± 0.11 mm) (Supplementary File 12). A photograph of the corresponding recording setup with the central food source is provided in Supplementary File 10.

## Locomotor Assay and Trajectory Analysis

To quantify locomotion independently of aggressive interactions, locomotion assays were conducted using the same arena and recording setup as the aggression experiments, but without a food source. For each assay, ten male flies of a single genotype (*Canton-S* or $w^{1118}$), aged 3–5 days post-eclosion, were placed into a transparent circular arena (6 diameter, 1.5 cm height) identical to that used in the aggression assays. Flies were allowed to freely explore the arena, and locomotion levels were recorded for approximately 2 min under either 25 or 18 °C. For each genotype, five independent videos were recorded at each temperature condition, resulting in five replicates per genotype per temperature.

Fly coordinates were extracted using a deep-learning-based tracking pipeline implemented with YOLOv10. Each individual fly was automatically detected and tracked in every video frame, yielding continuous x–y coordinate time series for each individual. Based on these coordinates, two locomotor indices were calculated: activity level and movement speed. Activity level was defined as the proportion of frames in which a fly exhibited detectable movement relative to the total number of frames, while movement speed was calculated as the total distance traveled divided by the total recording duration. These metrics and computational procedures are consistent with those used in our previous locomotion studies (Han et al. 2024). The trained model files are publicly available in an open-access repository (see Code Availability), and the extracted coordinate datasets for all individuals are provided in the Supplementary Materials (Supplementary File 11).

## Data Analysis

To obtain realistic body length values for scale representation in Fig. 1, the body length of male flies from each genotype was measured. At 3–5 days post-eclosion, ten *Canton-S* males and ten $w^{1118}$ males were randomly selected and imaged under a stereomicroscope at a fixed magnification. Images were calibrated using a stage micrometer under identical optical settings, and body length was defined as the linear distance from the head vertex to the abdominal tip. Measurements were performed using ImageJ software (NIH, USA) with a resolution of 0.01 mm. Individual body length values used for scale reference are provided in the Supplementary File 12.

To quantify offensive and escape actions in fruit flies, behavioral categories were defined based on established fly aggression ethograms (Chen et al. 2002; Nilsen et al. 2004) and refined according to our experimental observations (Table 1). For each trial conducted under the two assay temperature conditions (~25 °C and ~18 °C), the total number of occurrences of each behavioral event was recorded to assess condition-associated differences in behavioral frequency (Supplementary File 13). Supplementary File 13 provides the raw counts from all five annotators for each trial,

**Table 1** Behavioral categories and descriptions of offensive and escape actions

| Category | Behavior | Description |
|---|---|---|
| Offensive actions (non-contact) | Wing threat | One fly rapidly raises both wings to approximately 45° toward the opponent while facing it, without direct body contact |
| Offensive actions (physical contact–based) | Foreleg touching | One fly extends a single foreleg to briefly touch, tap, or push the opponent's leg or body, without sustained bilateral pushing or prolonged physical contact |
| | High-level fencing | One or both flies face each other and extend their forelegs forward to push against the opponent's body, involving bilateral or stronger foreleg pushing and more persistent physical contact than foreleg touching |
| | Holding | One fly grasps the opponent with its forelegs and attempts to restrain or immobilize the opponent for a short period, resulting in sustained body contact |
| | Chasing | One fly actively follows or attempts to drive the opponent away using forward body movement, without grasping or immobilization |
| Escape actions | Walking away | The fly turns away from the opponent and retreats without physical contact, typically following an aggressive interaction or threat |

interaction type, and behavioral category, together with the mean annotator count, the rounded final count used for analysis, and the trial-level calculations used to derive genotype-specific behavioral proportions.

Behavioral scoring was performed by manual annotation from video recordings. Automated classification tools, including JAABA, were evaluated but not adopted due to limited suitability for reliably distinguishing the specific set of offensive and escape behaviors examined in this study. Instead, all behaviors were independently annotated by five trained members of the author team using a frame-by-frame review approach. Annotators were blinded to both genotype and temperature condition during scoring, and no communication was allowed among annotators during the annotation process. For each behavioral metric, the final count was obtained by averaging the values across annotators and rounding to the nearest integer to minimize individual bias and scoring variability. Annotators were trained using a common behavioral definition guide prior to scoring.

The raw counts from all five annotators are provided in Supplementary File 13. For each trial, interaction type, and behavioral category, the final count used for analysis was calculated as the mean of the five annotator counts rounded to the nearest integer. Walking-away events were scored as independent escape events rather than as mandatory one-to-one responses to offensive actions, because some escape events occurred without clearly scored preceding offensive actions, whereas some offensive actions did not elicit observable escape responses. Therefore, offensive actions and walking-away events were analyzed as separate behavioral categories.

To compare relative behavioral contributions between genotypes, we further calculated the proportion of each behavior initiated by *Canton-S* versus $w^{1118}$ males during reciprocal intergenotype encounters. For these proportion-based analyses, only interactions in which *Canton-S* males acted toward $w^{1118}$ males and interactions in which $w^{1118}$ males acted toward *Canton-S* males were used. Same-genotype interactions, including *Canton-S* toward *Canton-S* and $w^{1118}$ toward $w^{1118}$, were included in the raw count data and in the analyses of absolute behavioral counts, but were not used for the proportion-based intergenotype analyses. For each behavior and each trial, the genotype-specific rates were calculated as follows:

$$Canton - S\,\text{rate}\,(\%) = \frac{Canton - S\,\text{toward}\,w^{1118}}{Canton - S\,\text{toward}\,w^{1118} + w^{1118}\text{toward}\,Canton - S} \times 100 \tag{1}$$

$$w^{1118}\,\text{rate}\,(\%) = \frac{w^{1118}\,\text{toward}\,Canton - S}{Canton - S\,\text{toward}\,w^{1118} + w^{1118}\text{toward}\,Canton - S} \times 100 \tag{2}$$

Therefore, the two rates were complementary and summed to 100% when the reciprocal intergenotype total was greater than zero. If the reciprocal intergenotype total was zero for a given behavior in a given trial, the corresponding rate was marked as NA and excluded from the statistical analysis for that behavior. The trial-level calculations are provided in Supplementary File 13.

Because the primary aim of this analysis was to determine which genotype exhibited greater behavioral initiative during direct competition, proportional measures were emphasized in the main analyses. Absolute counts of all offensive and escape behaviors for each genotype in every individual trial are provided in the Supplementary File 13 to facilitate reproducibility and independent comparison.

### Statistical Analysis

Behavior data were obtained from five independent test vials at 25 °C and three at 18 °C. Statistical analyses were performed using SPSS (version 22.0). Two types of behavioral data were analyzed: (1) count data, such as the number of offensive and escape actions, and (2) proportion data, such as the percentage of behaviors initiated by each genotype during pairwise interactions.

For count data, which are discrete and often deviate from a normal distribution, we used a Generalized Linear Mixed Model (GLMM) with a Poisson distribution and a log link function. In the model, Temperature Condition (25 °C vs. 18 °C) and Interaction Type (e.g., *Canton-S* vs. $w^{1118}$) were treated as fixed effects, while Trial number (five trials at 25 °C and three trials at 18 °C) was included as a random effect to account for repeated sampling. For each fixed effect, we report the estimated coefficient (*B*), standard error (*SE*), Wald chi-square statistic ($\chi^2$), associated *p*-value, and 95% confidence interval (*CI*). To aid interpretation, we additionally report the mean ± standard error of the mean (*SEM*) for each condition where relevant.

For proportion data (e.g., the relative contribution of *Canton-S* males to wing-based threat behaviors during intergenotypic encounters), the data were often highly skewed and not normally distributed. Therefore, we applied the non-parametric Mann–Whitney U test to compare assay-condition-associated differences. Unlike chi-square or Fisher's exact tests, which are designed for categorical frequency tables and may not perform well with small sample sizes or continuous proportion data, the Mann–Whitney U test allows for direct comparison of distributions without assuming normality. Moreover, because chi-square and Fisher's exact tests typically pool total counts across trials, they do not account for variation between independent replicates. In contrast, the Mann–Whitney U test treats each trial as an independent observation, allowing for more meaningful assessments across replicates. Furthermore, since our primary interest was to assess the relative dominance or initiative of each genotype, we focused on the proportion of actions rather than absolute counts. This emphasis on relative behavioral contribution further justifies the use of distribution-based non-parametric tests over count-based methods. Each behavioral metric was tested independently, and we report the *U*-value, *p*-value, mean ± standard error of the mean (*SEM*), and the effect size (rank-biserial correlation *r*).

Locomotion data (activity level and movement speed) were analyzed separately to evaluate assay-condition- and genotype-related differences in general movement capacity. For each video recording, locomotor metrics were first calculated for each individual fly and then averaged across the 10 males within the same vial, such that each vial (video) was treated as an independent replicate ($n = 5$ replicates per genotype per temperature). To test the effects of Temperature (18 °C vs. 25 °C), Genotype (*Canton-S* vs. $w^{1118}$), and their interaction on activity level and movement speed, two-way analysis of variance (ANOVA) was performed with both factors treated as between-subject variables. When significant main effects or interactions were detected, simple effects were examined to compare condition-associated differences within each genotype (~18 °C vs. ~25 °C) and genotype-related differences within each assay condition (*Canton-S* vs. $w^{1118}$). For ANOVA results, *F* values, degrees of freedom, *p* values, and partial eta squared (partial $\eta^2$) were reported. Locomotion data are presented as mean ± *SEM* based on vial-level replicates.

A *p*-value below 0.05 was considered statistically significant in all analyses.

## Results

### Genotype-related Differences in Offensive and Escape Actions Were Evident Under the Cooler Assay Condition

To examine how offensive and escape actions differed across assay conditions and genotypes, we quantified the total number of actions performed by each genotype toward both same-genotype and opposite-genotype opponents under the ~25 °C and ~18 °C assay conditions (Fig. 1). Offensive behaviors included physical and display-based actions, while escape behavior was measured by walking away responses (Table 1, Supplementary Files 1–8). Descriptively, *Canton-S* males showed more offensive actions under the ~25 °C assay condition than under the ~18 °C assay condition (25°C: mean ± *SEM* = 163.000 ± 38.548; 18°C: 32.000 ± 1.700, 5.093 times). For $w^{1118}$ males, the difference between assay conditions was even more pronounced, with values under the ~25 °C condition being 58.943 times higher than those under the ~18 °C condition (25°C: mean ± *SEM* = 157.200 ± 30.722; 18°C: 2.667 ± 0.981). The two genotypes presented highly similar levels of total offensive actions at 25°C. In terms of escape behavior, *Canton-S* males showed higher walking-away counts under the ~25 °C assay condition than under the ~18 °C assay condition (25°C: mean ± *SEM* = 3.600 ± 1.081; 18°C: 1.000 ± 0.471, 3.600 times), whereas $w^{1118}$ males showed a sharp decline in walking away behavior, with the 25 °C value being only 0.164 times that of 18 °C (25°C: mean ± *SEM* = 1.200 ± 0.716; 18°C: 7.333 ± 0.981).

To further compare the offensive behaviors of *Canton-S* males toward same-genotype and opposite-genotype opponents (Fig. 2A), we applied a Poisson Generalized Linear Mixed Model (GLMM) to statistically evaluate group-level differences in offensive action frequencies. The results showed that *Canton-S* males exhibited significantly more offensive actions toward $w^{1118}$ males at 25 °C than at 18 °C (25°C: mean ± *SEM* = 97.600 ± 18.910; 18°C: 7.333 ± 6.692; *B* = 90.267, *SE* = 20.059, Wald $\chi^2 = 20.251$, $p < 0.001$, 95% *CI* [48.867, 131.667]). A similar condition-associated difference was also observed in the number of offensive actions initiated toward same-genotype males, although the effect was weaker and did not reach statistical significance (25°C: mean ± *SEM* = 65.400 ± 15.479; 18°C: 24.667 ± 12.273; *B* = 40.733, *SE* = 19.754, Wald $\chi^2 = 4.252$, $p = 0.05$, 95% *CI* [−0.038, 81.505]). For $w^{1118}$ males, both types of offensive interactions showed significant differences between assay conditions. Offensive actions toward *Canton-S* males were markedly higher at 25 °C than at 18 °C (25°C: mean ± *SEM* = 89.200 ± 18.078; 18°C: 1.667 ± 3.190; *B* = 87.533, *SE* = 18.357, Wald $\chi^2 = 22.737$, $p < 0.001$, 95% *CI* [49.646, 125.421]). Similarly, offensive actions toward same-genotype males were also significantly higher under the ~25 °C assay condition than under the ~18 °C assay condition (25°C: mean ± *SEM* = 68.000 ± 15.784; 18°C: 1.000 ± 2.471; *B* = 67.000, *SE* = 15.976, Wald $\chi^2 = 17.588$, $p < 0.001$, 95% *CI* [34.026, 99.974]).

(a–b) Comparisons of the total number of offensive actions and escape actions (walking away) across four interaction types: *Canton-S* toward $w^{1118}$, *Canton-S* toward *Canton-S*, $w^{1118}$ toward *Canton-S*, and $w^{1118}$ toward $w^{1118}$, under 25 °C (warm) and 18 °C (cool) conditions. "*CS*" refers to *Canton-S*; "25°C" and "18°C" represent the two assay temperature conditions. Orange bars represent data collected at 25 °C, whereas blue bars represent data collected at 18 °C. Bars indicate mean values, and error bars represent the standard error of the mean (*SEM*). Each data point represents an independent trial (one arena per trial). Statistical significance is indicated directly in the figure (***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, $p < 0.1$ values are shown directly; Generalized Linear Mixed Model, GLMM).

In contrast, when escape behaviors were analyzed (Fig. 2B), differences between assay conditions were less consistent. For *Canton-S* males, walking away behavior toward $w^{1118}$ males was significantly higher under the ~25 °C assay condition than under the ~18 °C assay condition (25°C: mean ± *SEM* = 2.400 ± 0.748; 18°C: 0.333 ± 0.360; *B* = 2.067, *SE* = 0.830, Wald $\chi^2 = 6.201$, $p < 0.05$, 95% *CI* [0.353, 3.781]), while no significant variation was found toward same-genotype opponents (25°C: mean ± *SEM* = 1.200 ± 0.529; 18°C: 0.667 ± 0.509; *B* = 0.533, *SE* = 0.734, Wald $\chi^2 = 0.527$, $p = 0.475$, 95% *CI* [–0.982, 2.049]). For $w^{1118}$ males, escape actions when facing *Canton-S* males were significantly lower at 25 °C than at 18 °C (25°C: mean ± *SEM* = 0.800 ± 0.432; 18°C: 7.000 ± 1.650;

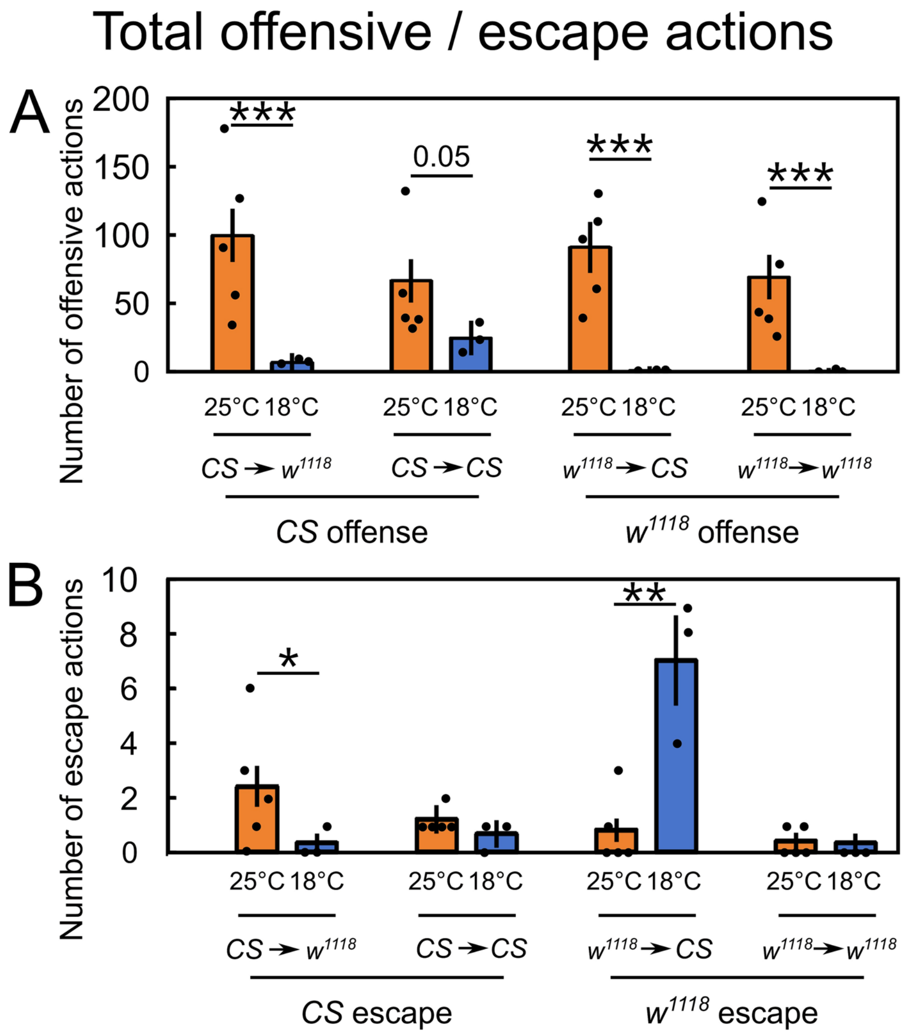


**Fig. 2** Offensive and escape actions in male fruit flies under two assay temperature conditions (**a**) totaloff ensive actions; (**b**) total escape actions (walking away)

$B = -6.200$, $SE = 1.706$, Wald $\chi^2 = 13.208$, $p < 0.01$, 95% *CI* [−9.720, −2.680]), whereas no significant condition-associated difference was observed in their walking away behavior toward same-genotype males (25°C: mean ± $SEM = 0.400 \pm 0.306$; 18°C: $0.333 \pm 0.360$; $B = 0.067$, $SE = 0.472$, Wald $\chi^2 = 0.020$, $p = 0.889$, 95% *CI* [−0.908, 1.041]). Although $w^{1118}$ males exhibited more frequent retreat behaviors at 18 °C, the overall number of such events remained low.

## Condition-associated Patterns of Wing Threat Behavior Between Male Fruit Fly Strains

Given that the total number of offensive and escape actions was markedly higher at 25 °C than at 18 °C, we focused our analysis on interactions between *Canton-S* and $w^{1118}$ males to compare the relative offensive and escape rates under the two assay conditions. Specifically, we calculated the rate of behaviors initiated by each genotype during intergenotype encounters and compared the relative contributions of *Canton-S* and $w^{1118}$ males across behavioral categories. As shown in Fig. 3, results are reported primarily for *Canton-S*. Because rates were calculated from reciprocal intergenotype encounters only, the $w^{1118}$ rate is complementary to the *Canton-S* rate for trials in which that behavior occurred; trials with no reciprocal intergenotype events for that behavior were treated as NA, as described in the Methods. However, this difference did not reach statistical significance ($U = 3.000$, $p = 0.169$, $r = -0.486$).

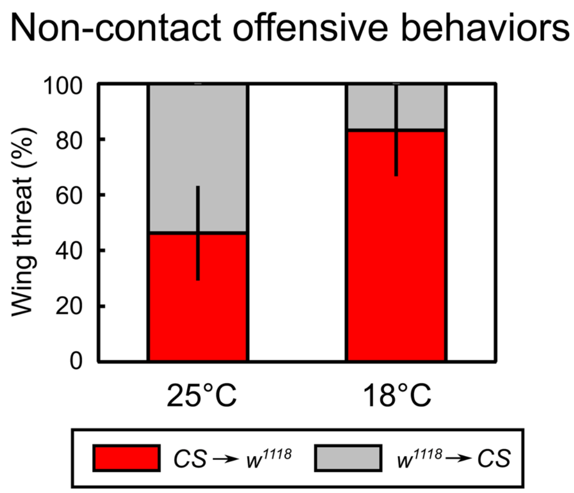


**Fig. 3** Wing threat behavior of different male fruit fly strains under two assay temperature conditions

The corresponding mean ± *SEM* values were 83.3% ± 16.7% at 18 °C and 46.2% ± 16.9% at 25 °C. Overall, these results indicate that wing threat behavior did not show a statistically significant condition-associated difference in the relative contribution of *Canton-S* and $w^{1118}$ males.

Bar plots showing the rates of wing threat initiated by *Canton-S* males toward $w^{1118}$ (red bars) and by $w^{1118}$ males toward *Canton-S* (grey bars) under warm (25 °C) and cool (18 °C) conditions. Wing threat represents non-contact wing-based exploratory or threatening behavior, as defined in Table 1. Group abbreviations follow those used in Fig. 2. Error bars indicate the standard error of the mean (SEM). Statistical comparisons were performed using the Mann–Whitney U test. Percentages were calculated from reciprocal intergenotype encounters only, using *Canton-S* toward $w^{1118}$ and $w^{1118}$ toward *Canton-S* counts. Trials with a reciprocal total count of zero were excluded from the corresponding proportion analysis.

### Condition-associated Patterns of Physical Contact-based Offensive Behaviors Between Male Fruit Fly Strains

To examine whether physical contact–based offensive behaviors showed condition-associated variation across genotypes, we analyzed the rates of foreleg touching, high-level fencing, holding, and chasing initiated by *Canton-S* and $w^{1118}$ males during direct competition (Fig. 4, Supplementary Files 4–7). As shown in Fig. 4A, *Canton-S* males exhibited a higher rate of foreleg touching at 18 °C (mean ± *SEM* = 100.0% ± 0.0%) than at 25 °C (mean ± *SEM* = 47.4% ± 4.7%), although this difference did not reach statistical significance ($U = 0.000$, $p = 0.102$, $r = -0.577$). For holding behavior (Fig. 4B), *Canton-S* males showed a significantly higher rate at 18 °C (mean ± *SEM* = 100.0% ± 0.0%) compared with 25 °C (mean ± *SEM* = 39.7% ± 4.8%; $U = 0.000$, $p < 0.05$, $r = -0.697$).

Similarly, the rate of high-level fencing was higher at 18 °C (mean ± *SEM* = 81.5% ± 9.8%) than at 25 °C (mean ± *SEM* = 51.4% ± 2.5%), showing a statistically significant difference ($U = 0.000$, $p < 0.05$, $r = -0.791$; Fig. 4C). The rate of chasing also tended to be higher at 18 °C (mean ± *SEM* = 100.0% ± 0.0%) than at 25 °C (mean ± *SEM* = 65.5% ± 5.1%), with the difference approaching the conventional threshold for statistical significance ($U = 0.000$, $p = 0.051$, $r = 0.691$; Fig. 4D). Together, these results indicate that, under cooler conditions, physical contact–based offensive behaviors were disproportionately initiated by *Canton-S* males during heterotypic interactions, whereas at 25 °C the relative contribution of the two genotypes to these behaviors was more comparable.

(a–d) Bar plots showing the percentages of foreleg touching, holding, high-level fencing, and chasing exhibited by *Canton-S* males toward $w^{1118}$ (red bars) and by $w^{1118}$ males toward *Canton-S* (grey bars). Abbreviations follow those used in Fig. 2. The error bars indicate the standard error of the mean. (*$p < 0.05$, $p < 0.1$ values are shown directly; Mann–Whitney U test). Percentages were calculated from reciprocal intergenotype encounters as described in Fig. 3.

### Relative Offensive and Escape Rates Between Male Fruit Fly Strains Across Assay Conditions

Furthermore, we summed all offensive behavioral metrics to calculate the offensive rate and used walking away as the measure of escape actions to calculate the escape rate (Supplementary File 8). The results showed that *Canton-S* males exhibited a clear advantage in terms of the offensive rate at 18 °C compared with 25 °C (18°C mean ± *SEM* = 80.8% ± 2.0%, 25 °C mean ± *SEM* = 50.5% ± 2.2%; $U = 0.000$, $p < 0.05$, $r = -0.791$; Fig. 5A). In contrast, $w^{1118}$ males exhibited more escape behavior at 18 °C, whereas

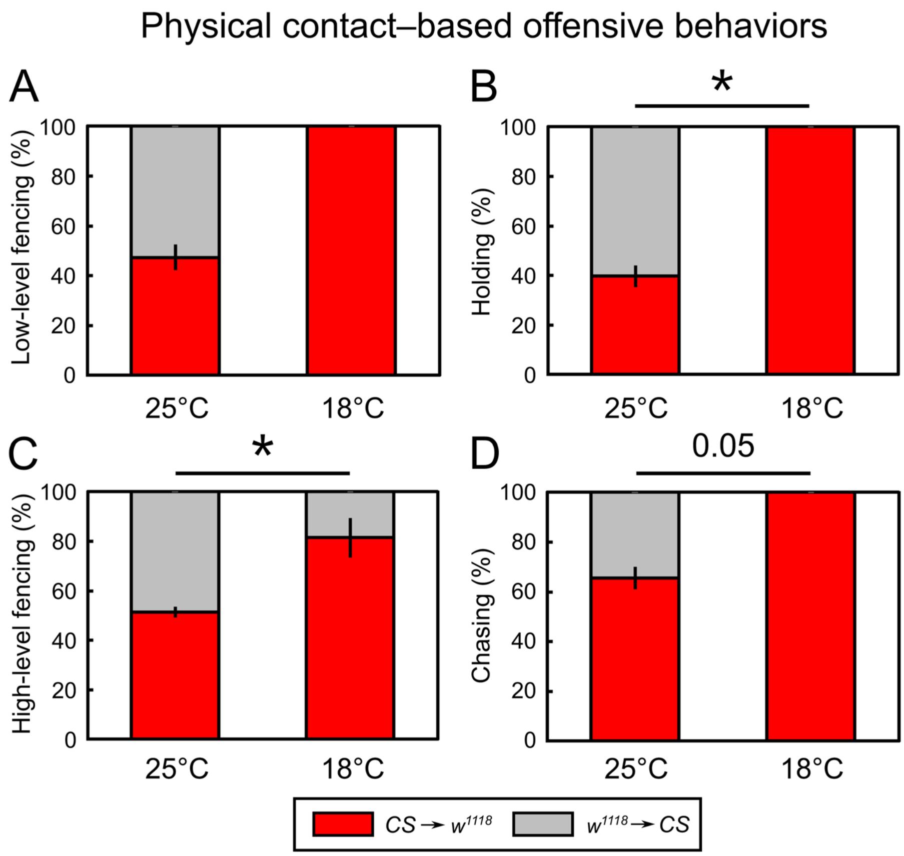


**Fig. 4** Physical contact–based offensive behaviors of different male fruit fly strains under two assay temperature conditions (**a**) foreleg touching; (**b**) holding; (**c**) high-level fencing; (**d**) chasing

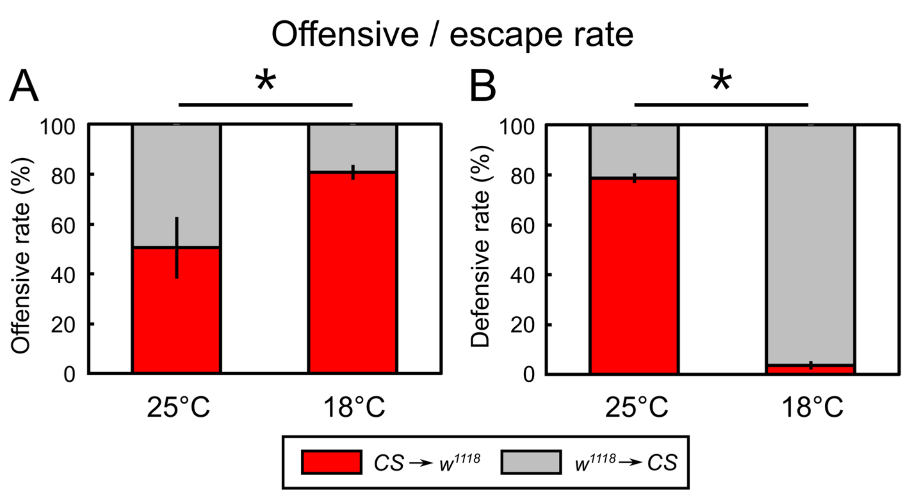


**Fig. 5** Relative offensive and escape rates of different male fruit fly strains under two assay temperature conditions (**a**) total offensive rate; (**b**) escape rate (walking away)

*Canton-S* males presented a higher escape rate at 25 °C (18°C mean $\pm$ *SEM* $=3.7\% \pm 3.7\%$, 25 °C mean $\pm$ *SEM* $=78.8\% \pm 12.7\%$; $U=0.000$, $p<0.05$, $r=-0.764$; Fig. 5B). Together, these findings suggest that genotype-related differences in relative offensive and escape rates were more evident under the ~18 °C assay condition, whereas the relative contribution of the two genotypes was more comparable under the ~25 °C assay condition.

(a–b) Bar plots showing the percentages of the total offensive rate and escape rate (walking away) exhibited by *Canton-S* males toward $w^{1118}$ (red bars) and by $w^{1118}$ males toward *Canton-S* (grey bars). Abbreviations follow those used in Fig. 2. The error bars indicate the standard error of the mean. (* $p < 0.05$; Mann–Whitney U test). Percentages were calculated from reciprocal intergenotype encounters as described in Fig. 3.

## Locomotion Under Different Assay Conditions and Genotypes

To examine whether general locomotion differed across assay conditions and genotypes, activity level and movement speed were analyzed separately using two-way analysis of variance (ANOVA) with Temperature (18 °C vs. 25 °C) and Genotype (*Canton-S* vs. $w^{1118}$) as between-subject factors (Fig. 6, Supplementary Files 9 and 11).

(a) Activity level and (b) movement speed of *Canton-S* and $w^{1118}$ males measured at 25 °C and 18 °C. Bars indicate mean values, and error bars represent the standard error of the mean (*SEM*). Each data point represents an independent trial (one arena per trial). Orange bars represent data collected at 25 °C, whereas blue bars represent data collected at 18 °C. Statistical significance is indicated directly in the figure (* $p < 0.05$; two-way ANOVA).

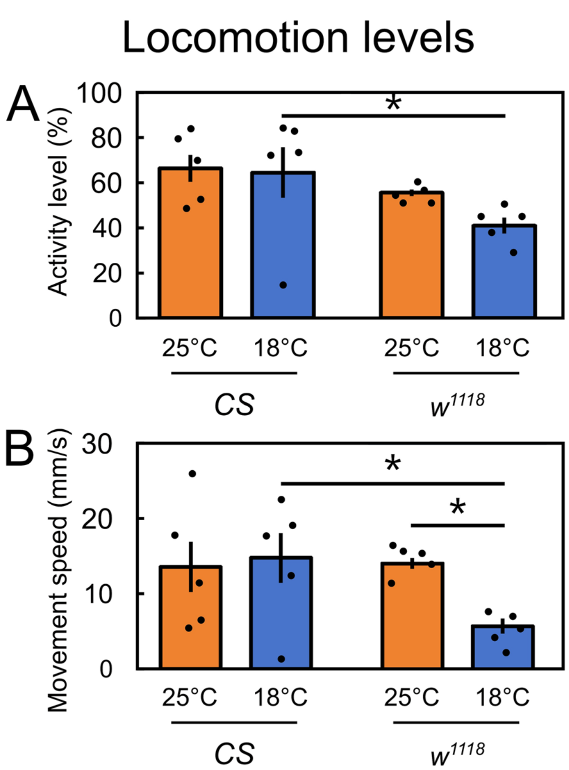


**Fig. 6** Locomotion of different male fruit fly strains under two assay temperature conditions (**a**) activity level; (**b**) movement speed

We first analyzed activity level between *Canton-S* and $w^{1118}$ males under different assay conditions (Fig. 6A). Overall, activity level differed between genotypes ($F(1, 16) = 5.418$, $p < 0.05$, partial $\eta^2 = 0.253$), whereas no overall difference was observed between assay conditions ($F(1, 16) = 1.242$, $p = 0.282$, partial $\eta^2 = 0.072$), and no interaction pattern between temperature and genotype was detected ($F(1, 16) = 0.750$, $p = 0.399$, partial $\eta^2 = 0.045$). Consistent with these overall effects, simple-effects analyses showed that at 18 °C, *Canton-S* males exhibited significantly higher activity levels than $w^{1118}$ males (*Canton-S*: mean ± *SEM* = 0.646 ± 0.125; $w^{1118}$: mean ± *SEM* = 0.409 ± 0.039; $p < 0.05$, 95% *CI* [0.014, 0.458]), whereas no genotype difference was observed at 25 °C (*Canton-S*: mean ± *SEM* = 0.664 ± 0.067; $w^{1118}$: mean ± *SEM* = 0.556 ± 0.016; $p = 0.317$, 95% *CI* [−0.114, 0.330]). Within each genotype, activity level did not differ significantly between 18 °C and 25 °C (*Canton-S*: $p = 0.863$, 95% *CI* [−0.240, 0.204]; $w^{1118}$: $p = 0.180$, 95% *CI* [−0.369, 0.075]).

Next, we analyzed movement speed between *Canton-S* and $w^{1118}$ males at 18 °C and 25 °C (Fig. 6B). Overall, no clear differences in movement speed were detected between genotypes ($F(1, 16) = 2.421$, $p = 0.139$, partial $\eta^{2 = 0.131)}$, or between temperature conditions (($F(1, 16) = 1.710$, $p = 0.209$, partial $\eta^2 = 0.097$), whereas the interaction between temperature and genotype approached the conventional threshold for statistical significance ($F(1, 16) = 3.060$, $p = 0.099$, partial $\eta^2 = 0.161$). Simple-effects analyses revealed that at 18 °C, *Canton-S* males moved significantly faster than $w^{1118}$ males (*Canton-S*: mean ± *SEM* = 14.785 ± 3.716; $w^{1118}$: mean ± *SEM* = 5.763 ± 1.101; $p < 0.05$, 95% *CI* [0.839, 17.204]), whereas no genotype difference was observed at 25 °C (*Canton-S*: mean ± *SEM* = 13.580 ± 3.757; $w^{1118}$: mean ± *SEM* = 14.108 ± 0.812; $p = 0.893$, 95% *CI* [−8.710, 7.655]). In addition, within the $w^{1118}$ genotype, movement speed was significantly lower at 18 °C than at 25 °C ($p < 0.05$, 95% *CI* [−16.526, −0.161]), whereas no condition-associated difference was detected in *Canton-S* males ($p = 0.759$, 95% *CI* [−6.978, 9.388]).

Together, these results indicate that differences in locomotion between *Canton-S* and $w^{1118}$ males are most evident under the cooler assay condition, whereas locomotion is largely comparable between genotypes at 25 °C.

## Discussion

In the present study, we examined offensive and escape actions of two commonly used male fruit fly strains under two assay temperature conditions approximating 25 °C and 18 °C. Overall, offensive actions were more frequent under the ~25 °C assay condition than under the ~18 °C assay condition. However, genotype-related differences were most evident under the cooler assay condition: *Canton-S* males maintained higher levels of offensive behavior than $w^{1118}$ males at ~18 °C, whereas the two genotypes showed more comparable offensive levels at ~25 °C. Baseline locomotor assays further showed genotype-related differences under the cooler condition, with *Canton-S* males exhibiting higher activity levels and movement speed than $w^{1118}$ males at ~18 °C. These findings suggest that food competition-related behaviors differ between *Canton-S* and $w^{1118}$ males across assay conditions, and that strain differences are most pronounced under cooler conditions.

Although all flies were cultured under identical laboratory conditions, the two behavioral assay conditions were conducted during different months; therefore, the observed differences between the ~18 °C and ~25 °C assay conditions cannot be attributed exclusively to assay temperature. Nevertheless, because the assays were conducted indoors, temperature was actively maintained within a narrow range, and flies were reared under a fixed photoperiod, the present design still allows us to describe condition-associated behavioral patterns and genotype-related differences under cooler and warmer assay conditions. The higher offensive counts observed under the ~25 °C assay condition are broadly consistent with previous studies showing strong temperature dependence of insect physiology and behavior (Schou et al. 2013; Bahrndorff et al. 2016; Hannigan et al. 2023). Previous work has shown that lower temperatures are generally associated with reduced activity levels in fruit flies and may induce diapause-like physiological states, whereas extremely low temperatures can lead to chill coma or impaired survival (Jensen et al. 2007; Anduaga et al. 2018). Therefore, the lower offensive counts observed under the ~18 °C assay condition may reflect constrained behavioral performance under cooler conditions, consistent with thermal performance curves in ectothermic animals. However, because the two assay conditions were conducted during different months, this interpretation should be considered condition-associated rather than a definitive isolated temperature effect. Our locomotor assays further showed that genotype-related differences in locomotion were evident under the cooler condition, with movement speed being lower in $w^{1118}$ males than in *Canton-S* males at ~18 °C, whereas no consistent condition-associated difference in overall activity level was detected. Thus, the lower offensive behavior observed in $w^{1118}$ males under the cooler condition cannot be explained simply by a uniform decline in general locomotion.

Under the ~25 °C assay condition, $w^{1118}$ males showed offensive behavior levels comparable to *Canton-S* males, whereas under the ~18 °C assay condition they exhibited substantially lower offensive behavior than *Canton-S* males. This pattern indicates genotype-specific differences in the thermal sensitivity of aggressive behavior. Mutations in the *white* gene have been shown to influence multiple physiological processes beyond eye pigmentation, including neural and metabolic function (Qiu et al. 2017), and $w^{1118}$ flies display altered stress tolerance under certain environmental conditions (Hoffmann and Parsons 1989a, b; Qiu et al. 2017). These physiological differences may contribute to variation in the thermal sensitivity of aggressive behavior across genotypes, without necessarily producing parallel changes in baseline locomotion. Notably, *Canton-S* males maintained relatively high levels of offensive behavior under the cooler assay condition in both heterotypic and homotypic interactions, suggesting reduced behavioral sensitivity to cooling in this strain. Overall, the observed genotype differences in aggression at 18 °C are consistent with genotype-dependent thermal sensitivity of behavioral performance, rather than differences in general locomotion level per se.

Interestingly, when comparing escape behavior across social contexts under the cooler assay condition, $w^{1118}$ males exhibited relatively higher levels of escape behavior toward *Canton-S* males, while their escape responses toward conspecifics ($w^{1118}$ vs. $w^{1118}$) were notably lower (Fig. 2b). Although the absolute difference in frequency was modest (an average of approximately seven additional responses), this pattern suggests that escape behavior in $w^{1118}$ males was selectively expressed in heterotypic interactions under the cooler assay condition. Consistent with this interpretation, $w^{1118}$ males experienced more attacks from *Canton-S* males than from conspecifics under the cooler assay condition (Fig. 2a). Taken together, these results indicate that escape responses in $w^{1118}$ males were not uniformly elevated under the cooler assay condition, but instead depended on the social context of the interaction. Importantly, because escape behavior emerges from direct social interactions, the elevated escape rate of $w^{1118}$ males in heterotypic encounters at 18 °C likely reflects not only intrinsic properties of $w^{1118}$ males, but also differences in the thermal sensitivity of behavioral performance between the two genotypes. In particular, *Canton-S* males maintained relatively high levels of offensive behavior under the cooler assay condition, which may have increased the likelihood of eliciting escape responses from $w^{1118}$ males that are more behaviorally constrained by cooling. These findings further support the view that social behaviors varied across assay conditions in a context- and genotype-related manner, rather than through a generalized increase in escape tendency or locomotion. In the present study, we focused on characterizing behavioral differences in offensive and escape responses between genotypes across assay conditions, without addressing the underlying mechanisms. Future studies incorporating direct measurements of energy reserves, metabolic rates, and survival across assay conditions will be necessary to determine whether these context-dependent escape patterns reflect differential thermal sensitivity of behavioral performance or involve additional physiological constraints.

A methodological limitation of the present study is that the behavioral arena did not strictly confine flies to a two-dimensional monolayer. Previous fly behavioral studies have used shallow physical chamber designs or chemical barriers to reduce vertical movement and prevent flies from walking on the lid or side wall (Simon and Dickinson 2010; Wang and Anderson 2010). In contrast, our assay used a 1.5 cm-high transparent arena with a clear lid. This design was chosen because the assay involved group-level competition around a central spherical food source, and an overly shallow chamber could have interfered with flies standing on or around the food, occupying the food patch, and performing contact or withdrawal behaviors near the food source. Nevertheless, this design did not completely prevent vertical displacement, including occasional walking on the side wall or underside of the lid. Such movement could reduce the visibility of some interactions and introduce variability into both locomotor measurements and aggression counts. Although ambiguous or occluded events were excluded from scoring, comparisons with studies using strict monolayer arenas or automated tracking systems should be made cautiously. Future studies should use a lower chamber height, a sloped-wall or shallow physical design, or a validated chemical barrier to better restrict flies to a monolayer while preserving access to the food source. Multi-angle recording could also help distinguish vertical movement from true locomotor and social interaction events. In addition, because brief cold immobilization was used for transfer, short-term handling effects cannot be completely excluded, although the same transfer procedure and recovery interval were applied to all genotypes and assay conditions. Future studies could further minimize handling effects by using aspiration-based transfer or longer standardized acclimation periods before recording.

While assay temperature was the intended primary difference between the two testing conditions, the experiments were conducted in different seasons, during which other environmental parameters may also have differed. Although all behavioral assays were performed indoors and flies were reared in incubators under a fixed photoperiod, thereby minimizing potential effects of seasonal variation in day length, ambient humidity and atmospheric pressure were not independently controlled. At the study location, during April–May, relative humidity typically remains high, generally ranging from approximately 80%–95%, while atmospheric pressure is comparatively lower and more variable, typically ranging from ~1005 to 1012 hPa. In contrast, during November–December, relative humidity is usually lower, commonly ranging from ~40%–70%, with occasional drops below ~45%, whereas atmospheric pressure tends to be higher and more stable, typically ranging from ~1018 to 1025 hPa. These environmental parameters covary with seasonal temperature changes and could potentially influence physiological state or behavioral responsiveness. However, because humidity and barometric pressure were neither experimentally manipulated nor systematically recorded, their specific contributions cannot be disentangled from the assay temperature conditions in the present study. Accordingly, our interpretations focus on condition-associated behavioral patterns and genotype-related differences within each assay condition, while acknowledging that future studies using fully controlled environmental chambers will be required to isolate the specific effects of temperature, humidity, and atmospheric pressure.

It should be noted that same-strain males also exhibited clear offensive and escape behaviors during resource competition, as reflected in our data. In our experiments, we not only observed interactions between *Canton-S* and $w^{1118}$, but also recorded behaviors between *Canton-S* males and between $w^{1118}$ males (as shown in Fig. 2). The results in Fig. 2 show that, when comparing interactions within the same genotype within the same assay condition, males of both strains engaged in offensive and escape behaviors during food competition, with these interactions occurring more frequently under the ~25 °C assay condition. Although same-strain interactions were therefore present across assay conditions, the primary focus of the present study was the comparison of behavioral responses between genotypes across assay conditions, particularly during heterotypic interactions. This experimental design more clearly reveals relative differences in behavioral responsiveness between strains and minimizes potential masking effects arising from social familiarity among individuals of the same genotype. Nevertheless, resource competition also occurs between same-strain males, and this has been clearly supported by our data. These results indicate that aggressive interactions varied across assay conditions and social contexts, but the contribution of temperature per se should be tested further under fully controlled environmental conditions.

Another notable pattern emerged when comparing different categories of offensive behavior between genotypes under the cooler assay condition. Strain-related differences were less pronounced in wing-based threat behaviors than in physical contact–based behaviors. Specifically, under cooler conditions (18 °C), $w^{1118}$ males showed a marked reduction in contact-based offensive behaviors relative to *Canton-S* males, whereas wing threat behaviors remained relatively similar between the two genotypes. This dissociation suggests that different components of aggressive behavior may differ in their sensitivity to condition-associated physiological constraints on behavioral performance. Wing threat behaviors typically involve brief, low-intensity movements without sustained physical contact, whereas contact-based behaviors require greater biomechanical engagement and continuous behavioral execution. Consequently, under the cooler assay condition, behaviors involving direct physical contact may be more strongly associated with constraints on behavioral performance, leading to more pronounced genotype differences in these behaviors. In contrast, wing-based threat behaviors appear comparatively robust to cooling, resulting in weaker strain differences across assay conditions. Overall, this pattern is consistent with the possibility that behavioral modules differ in their sensitivity to cooler assay condition.

In this study, food competition-related offensive and escape behaviors differed between the ~18 °C and ~25 °C assay conditions, with genotype-related differences being most evident under the cooler condition. *Canton-S* males maintained higher levels of offensive behavior and locomotor performance than $w^{1118}$ males under the ~18 °C condition, whereas the two genotypes were more comparable under the ~25 °C condition. These findings suggest that genotype-related differences in competitive behavior are more pronounced under the cooler assay condition and may be associated with differences in behavioral performance and baseline locomotion. Because the two assay conditions were conducted during different months and humidity and barometric pressure were not independently controlled, differences between the ~18 °C and ~25 °C conditions should be interpreted as condition-associated patterns rather than definitive isolated temperature effects. Future studies using fully controlled environmental chambers will be needed to separate the effects of temperature from other atmospheric variables and to clarify the neural and physiological mechanisms underlying genotype-dependent variation in food competition-related social behavior.

**Acknowledgements** We are grateful to all the lab members for their suggestions and collaboration in this study. We also thank the National Demonstration Center for Experimental Life Sciences Education (Xiamen University) for providing the experimental environment and fruit fly.

**Author Contributions** R.H. conceived and designed the study, developed the methodology, performed formal analyses, acquired funding, and supervised the project. J.Z., J.-X.W., Y.-X.W., A.-Q.W., H.-W.W., X.L., X.-H.L., L.-X.L., and K.-J.F. carried out material preparation, data collection, and data analysis under the supervision of R.H. R.H. and J.-X.W. wrote the first draft of the manuscript. All authors contributed to manuscript revision, commented on previous versions of the manuscript, and approved the final version.

**Funding** This work was supported by the National Natural Science Foundation of China (Grant No. 32500921); the Fujian Provincial Natural Science Foundation of China (Grant No. 2024J08018); the Zhangzhou Natural Science Foundation Project (Grant No. ZZ2024J07); and the Young and Middle-aged Teachers' Education and Scientific Research Projects in Fujian Province (Science and Technology Category) (Grant No. JAT220508).

**Data Availability** The data supporting the results of this study are available in the article and its supplementary materials.

**Code Availability** The code is available at https://github.com/sguistar/FlyDetection.

**Declarations**

**Competing interests** The authors declare no competing interests.

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